

Appendix A2: conditional probability tables.

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Network structure

Decision nodes

Controls the climate and fishing management scenarios, those are defined a priori depending on the conditions to be explored in the scenario analyses.

Extreme weather events (node: event)

The node *Event* is used to wrap all the extreme weather events we are considering.

Levels (and acronyms)

- **Storms** (sto). Strong wind events, with intensity above 24.5 m/s. Season: summer.
- **Gales** (gal), or moderately strong wind events, with intensity between 15 m/s and 24.5 m/s;
- **Extreme precipitation** (erf) – rainstorm (summer), or deviation from average precipitation regime;
- **Algal bloom** (abl): intense algal bloom;
- **Summer heatwaves** (hws): prolonged discrete anomalously warm water event (Hobday et al., 2016);
- **Winter heatwaves** (hww): prolonged discrete anomalously warm water event (Hobday et al., 2016);
- **Icing** (Ici): strong wind events associated to ice, which then forms icing pile up on gears and boats.

Fishing style

The node *Fishing style* is used to wrap all the fishing styles and seasons.

Levels (and acronyms)

- **Recreational fisher in winter season** (rew): a fishing style that primarily performs ice fishing and potentially can fish in open water from the shore;
- **Recreational fisher in summer season** (res): a fishing style that primarily performs ice fishing and potentially can fish in open water from the shore;
- **fishing guide in winter season** (fgw), a fishing style that primarily performs ice fishing and potentially can fish in open water from the shore;
- **fishing guide in the summer season** (fgw), which primarily fish in coastal areas from shore or boat and can potentially move to inland fishing;
- **Coastal small-scale fishers**, (ssf) a fishing style that happen mostly in summer using gillnets in coastal areas, and in some cases with traps (not for salmon) but can perform limited amount of ice fishing in winter;
- **Coastal small-scale fishers specialised in salmon fishing** (sal): which fish for salmon with fixed traps rigidly regulated;
- **Pelagic trawlers** (trw), represented by pelagic trawlers mostly active in winter and exclusively fishing for small pelagic;
- **Archipelago fishers** (arc), which is similar to an artisanal fishers but is by law prevented from selling their catches.

Fishing management

it controls how much fishing regulations allows fishers to implement adaptation strategies (e.g.: changing target species and gear).

Levels (and acronyms)

- **Business as Usual** (BAU): each fishing style adopts strategies and behaviour according to interviews results
- **Flexible management** (FLEX). Fishing style is allowed for more flexibility.

Stock status

it controls how much fishing regulations allows fishers to implement adaptation strategies (e.g.: changing target species and gear).

Levels (and acronyms)

- **status quo** (SQ): each fishing style adopts strategies and behaviour according to interviews results
- **improved stock status** (FLEX). Fishing style is allowed for more flexibility.
- **worsened stock status**

Chance nodes

Infrastructure

Parents: fishing_style.

Levels: no, some, extensive.

Description: availability of terrestrial infrastructures such as roads, ports and bridges. CPT is already filled with method M1 based on the answer to Q17 (infrastructure).

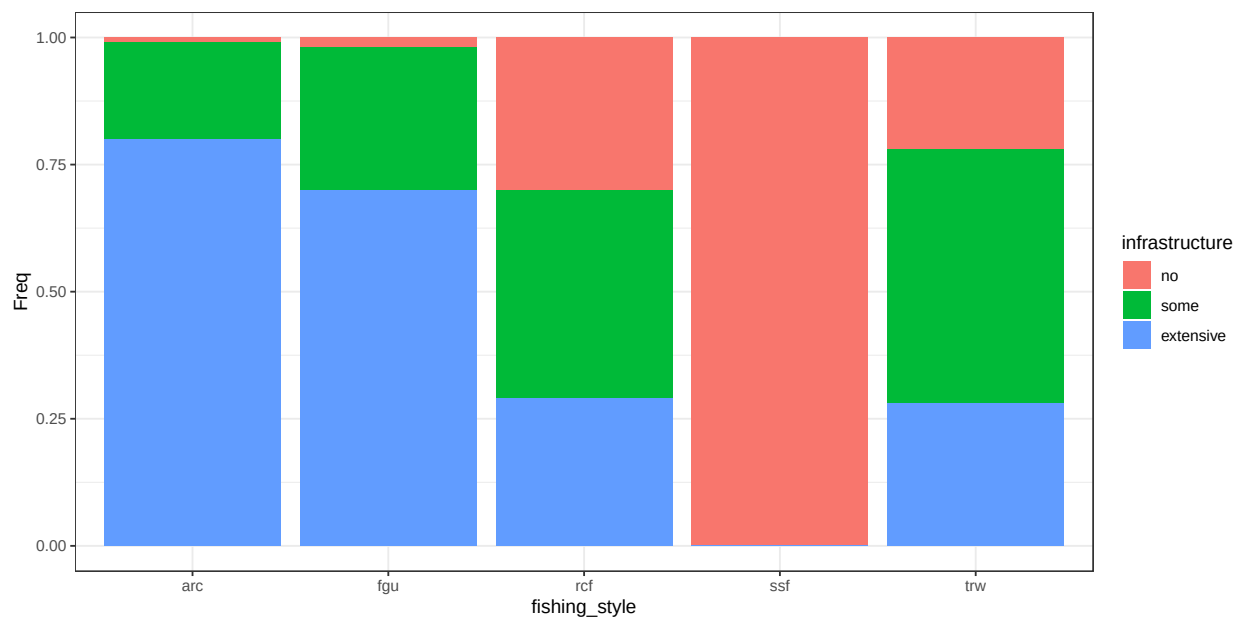


Figure A2.1: conditional distribution for infrastructure .

Flexibility

Parents: management, infrastructure.

Levels: rigid, status_quo, flexible.

Description: choosing different flexibility possibility to change fishing strategy (both gear and target). This can be linked to the node “strategy”.

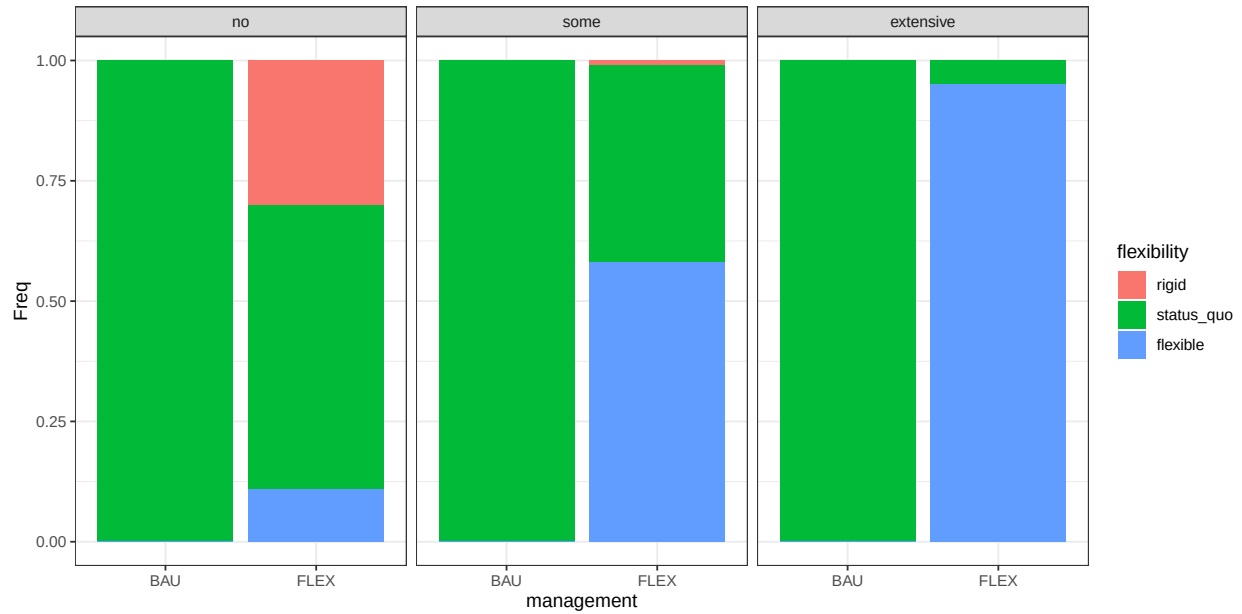


Figure A2.2: conditional distribution for flexibility . Columns represents levels of the node infrastructure.

Strategy

Parents: fishing_style, event, flexibility.

Levels: cope, react, adapt, not_relevant.

Description: Fishers response is coded as Adapt (proactive planning); react (unplanned response); cope (accept passively) (Green et al., 2021). Cope is when fisher prefer to not go out fishing; React is when they apply small modification to their practice, such as fishing in deeper areas or in different time of the day; Adapt is when they take proactive response such as modify their strategy by changing fishing target and/or gear..

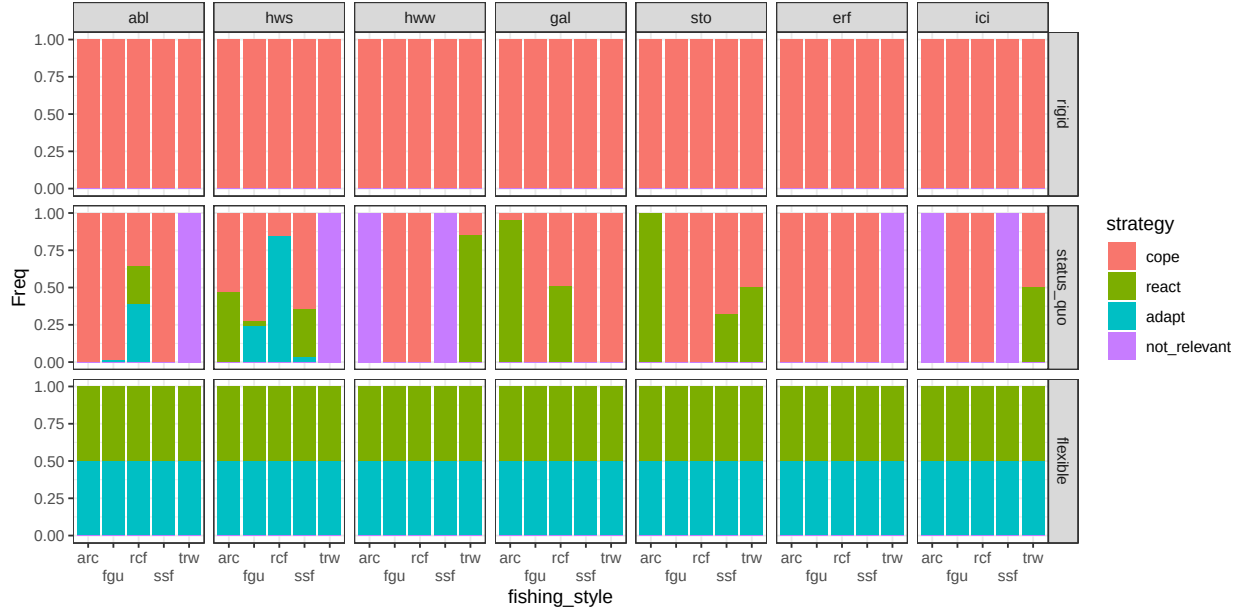


Figure A2.3: conditional distribution for strategy . Columns represents levels of the node event; rows are levels of the node flexibility.

Additional mitigation

Parents: strategy, fishing_style, event.

Levels: no, travel_further, short_sets, other_technical.

Description: Describes the measures implemented in the case when the strategy is React. The different solutions work differently and have different cost: travel further implies fisherman change fishing location to seek a sheltered place or to find deeper waters; short sets mean that the fisher perform shorter sessions than average (i.e.: reduced soak time for gillnets); other technical includes strategies that are mentioned sporadically (i.e.: use of ice machines). The CPT is compiled manually for each combination of event and fisherman based on what has been mentioned throughout the questionnaire. In case only one strategy is mentioned, this is assigned probability = 1. When two strategies are mentioned, they are 50% probable unless one strategy is explicitly adopted more often than the other (and so on for three strategies)..

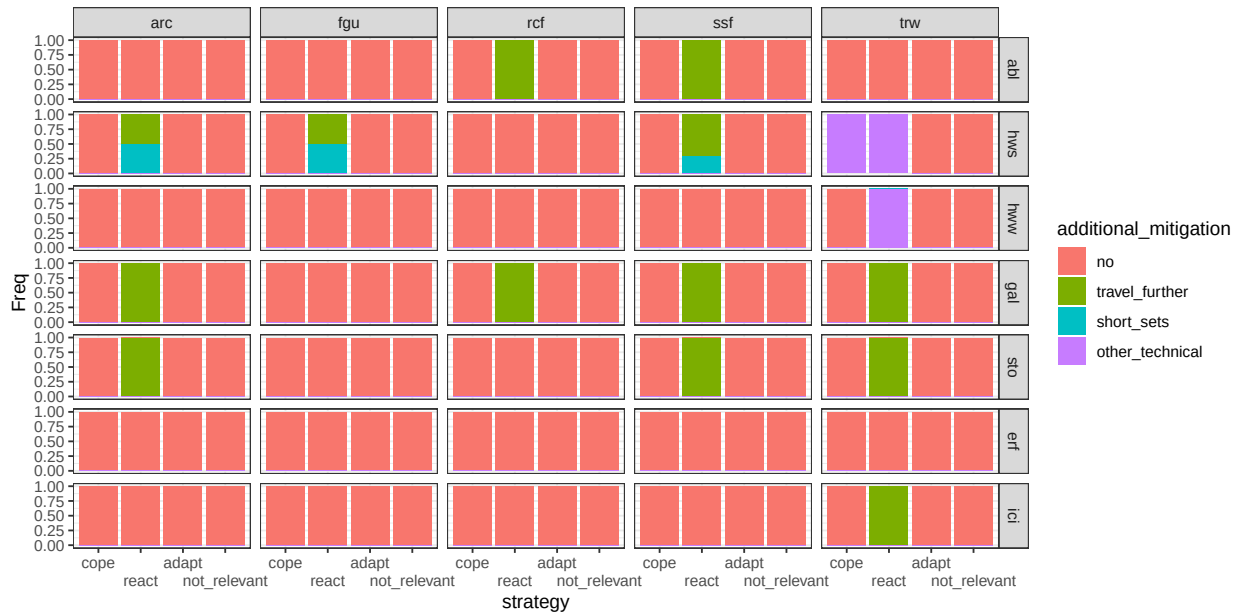


Figure A2.4: conditional distribution for `additional_mitigation`. Columns represents levels of the node `fishing_style`; rows are levels of the node `event`.

Gear

Parents: fishing_style, event, strategy.

Levels: otm, gns_spf, gns_fws, fpn_ana, rod, icefishing, inland, not_relevant.

Description: Describes the most probable gear used by each fisherman. The CPT is compiled manually for each combination of event and fisherman. In the case when the strategy is cope or react, the CPT reflects what we do expect from the fisher in normal conditions (this can be one or more gears). When the strategy is Adapt, the CPT accounts for what has been mentioned throughout the questionnaire..

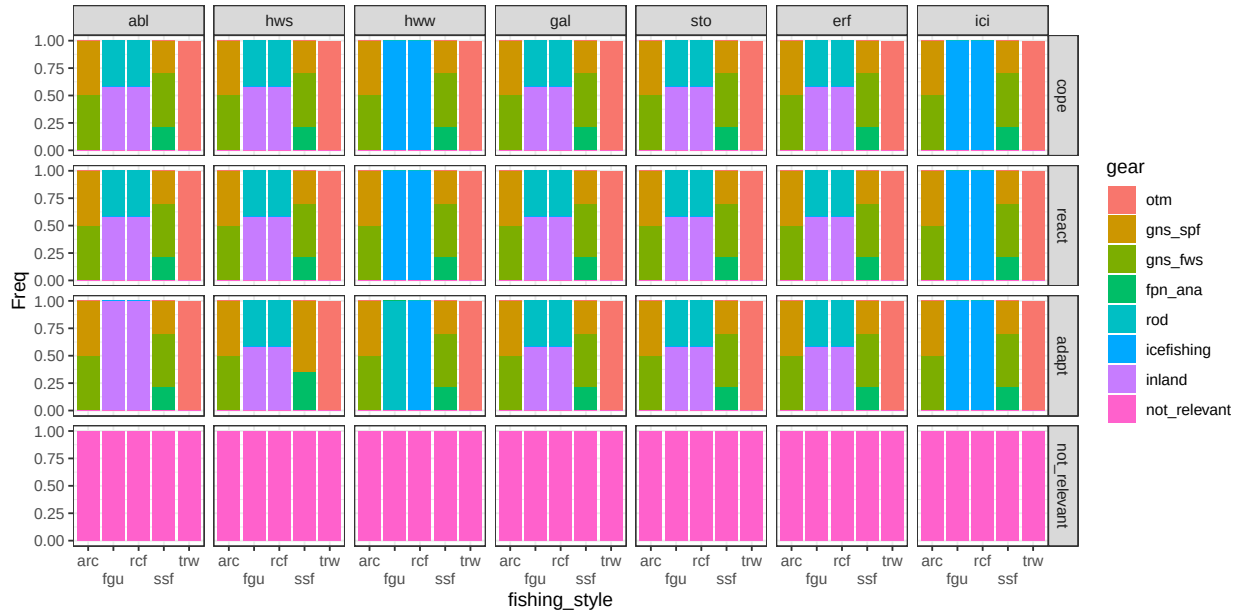


Figure A2.5: conditional distribution for gear . Columns represents levels of the node event; rows are levels of the node strategy.

Catchability

Parents: target, event, gear.

Levels: worse, same, better.

Description: the ability to catch fish depending on gear-fish interactions. For example, a gillnet clogged in algae is more visible and less efficient (less catchability); a gillnet in murky waters is less visible and more efficient (more catchability); the necessity to go to haul the net sooner because of heatwave diminish the soaking time (less catchability). For each combination of species and even the CPT is filled with method M1-bis based on the answer to Q3 (catchability)..

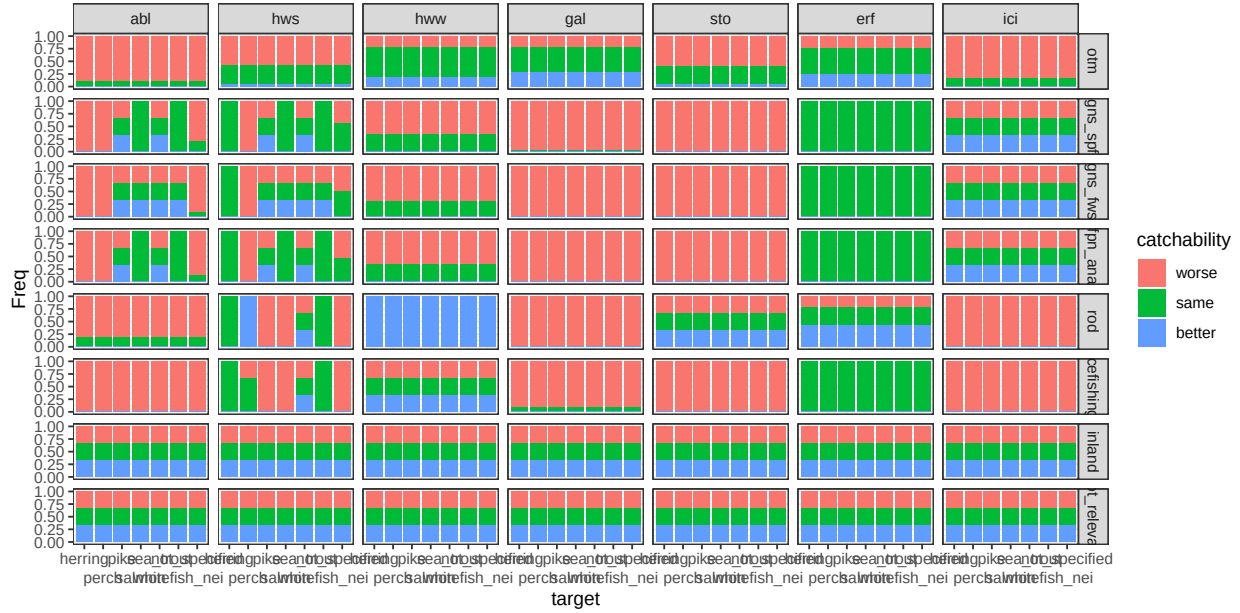


Figure A2.6: conditional distribution for catchability . Columns represents levels of the node event; rows are levels of the node gear.

Personal safety

Parents: event, gear, additional_mitigation.

Levels: no, minor, major.

Description: any variation to ordinary working condition that can increase the potential physical damage for operators (i.e.: icing can require more time on the deck that translates in increased risk of slipping; a violent storm is life threatening). When the technical solution is no, the CPT is filled with method M1-bis based on the answer to Q9 (damage). When the technical solution is “travel further”, the personal safety is “no” with probability 1. With the available answers no other measure is implemented to deal with personal safety..

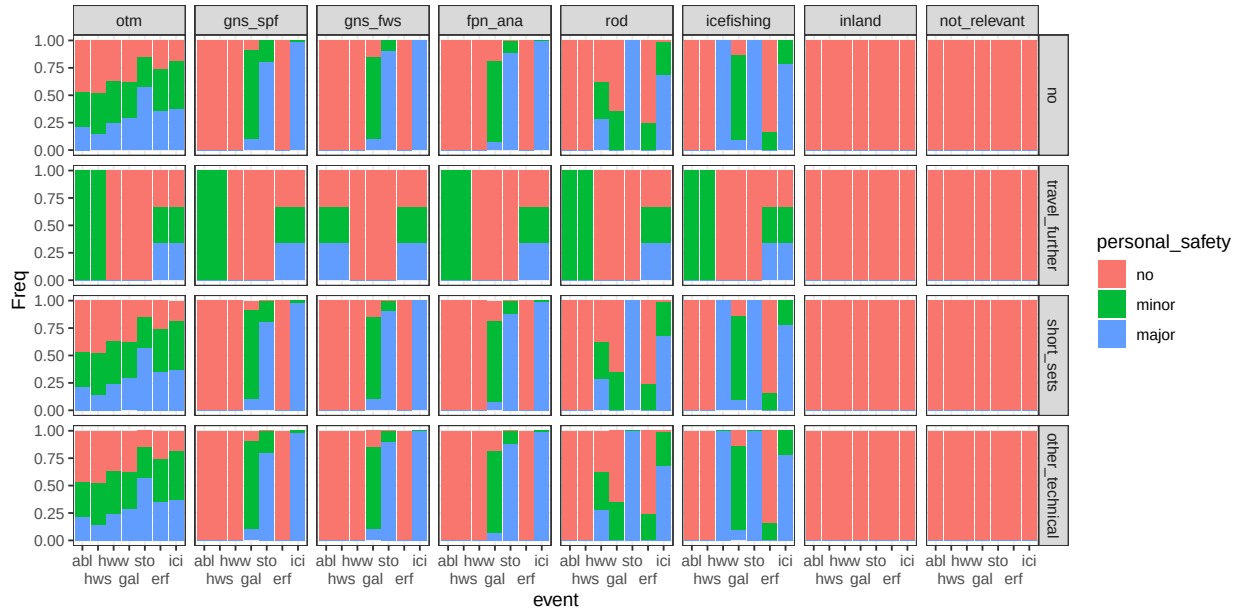


Figure A2.8: conditional distribution for `personal_safety`. Columns represents levels of the node `gear`; rows are levels of the node `additional_mitigation`.

Damage

Parents: event, gear, additional_mitigation.

Levels: no, minor, major, destroy.

Description: damage to the fishing gear ranging from minor to completely losing the fishing gear or part of it because it was lost or irreparably damaged. When the technical solution is no, the CPT is filled with method M1-bis based on the answer to Q10 (damage). When the technical solution is “travel further”, the damage is “no” with probability 1. With the available answers no other measure is implemented to deal with damages..

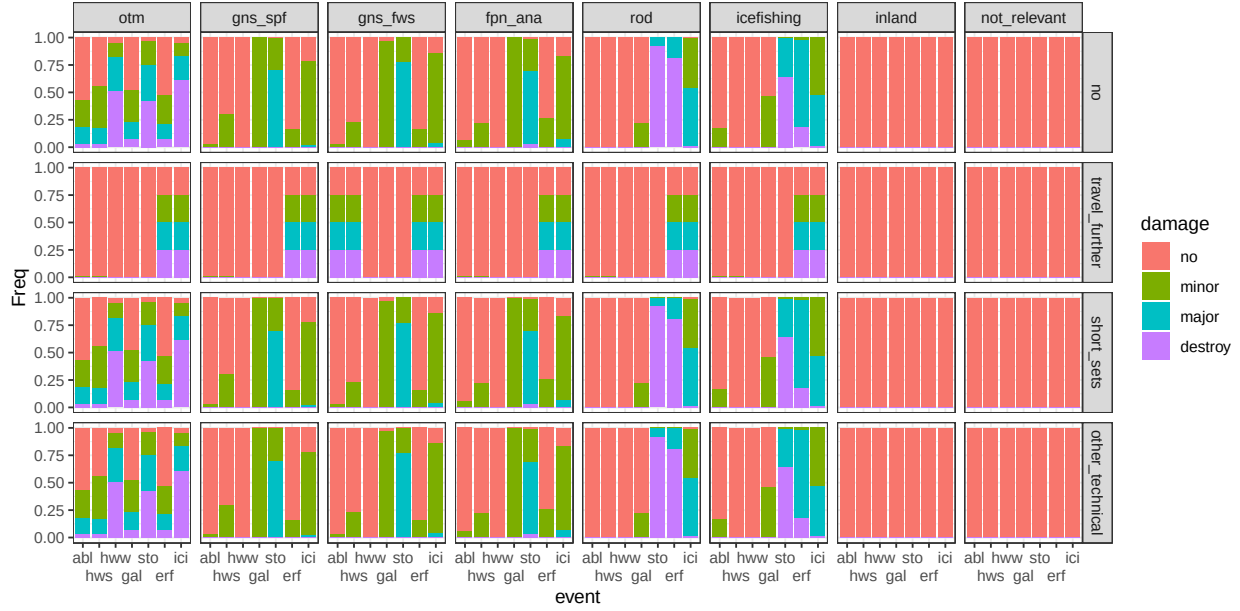


Figure A2.9: conditional distribution for damage . Columns represents levels of the node gear; rows are levels of the node additional_mitigation.

Target

Parents: gear, strategy, event.

Levels: herring, perch, pike, salmon, sea_trout, whitefish_nei, not_specified.

Description: Describes the most probable target species associated to each fishing gear. The CPT is compiled manually for each combination of event and gear. In the case when the strategy is cope or react, the CPT reflects what we expect from the fisher in normal conditions (this can be one or more combinations of gear/species). When the strategy is Adapt, the CPT accounts for what has been mentioned throughout the questionnaire. Example: gillnetters normally target a set of species (herring, whitefish, perch), but when there is an heatwave they switch primarily to perch..

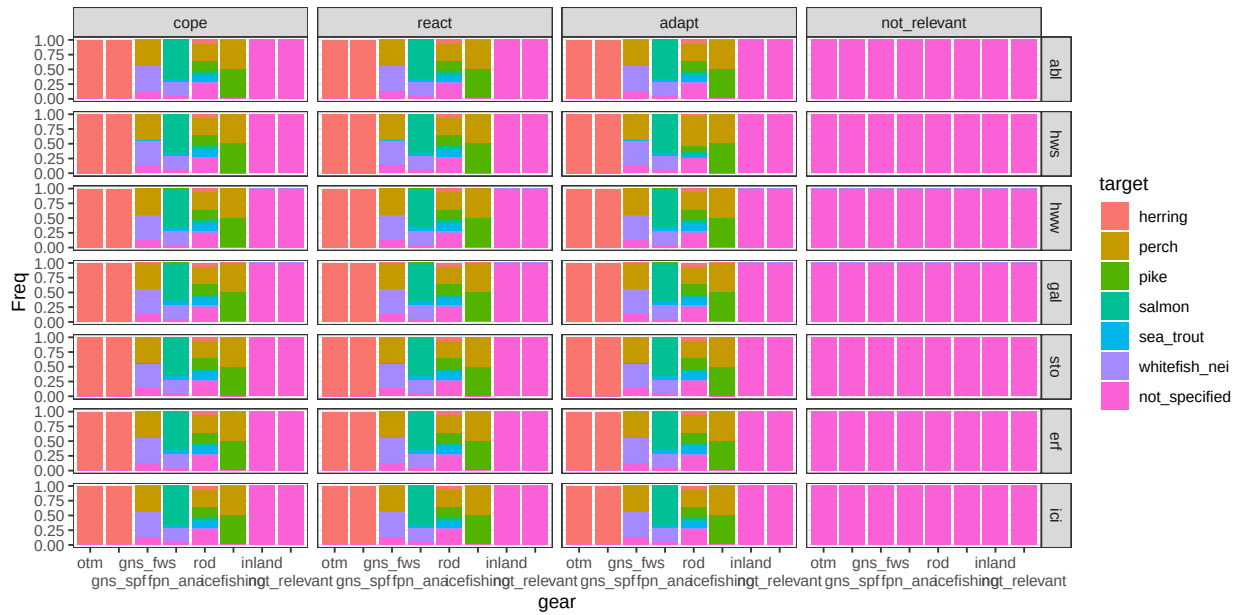


Figure A2.10: conditional distribution for target . Columns represents levels of the node strategy; rows are levels of the node event.

Go out

Parents: event, fishing_style, strategy.

Levels: no, yes, not_relevant.

Description: Describes the probability to go out fishing. When the parent node strategy is “cope”, the CPT is filled with method M1 based on the answer to Q2 (go_out). In the case when the parent node strategy is “react” or “adapt”, the decision is always to go fishing..

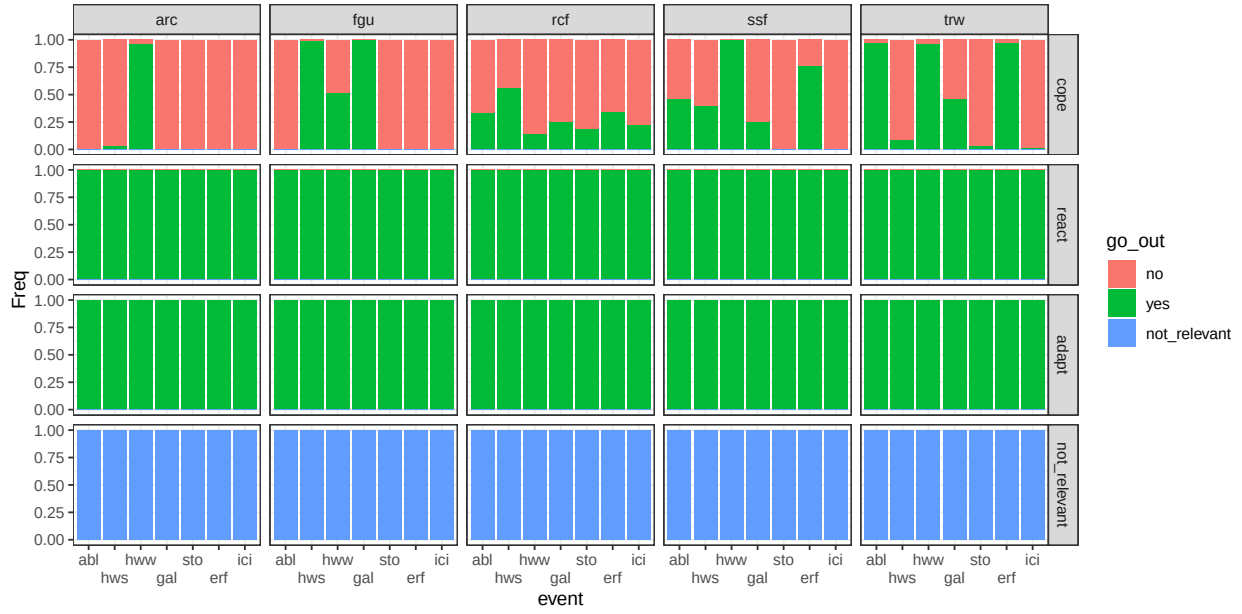


Figure A2.11: conditional distribution for go_out . Columns represents levels of the node fishing_style; rows are levels of the node strategy.

Stress

Parents: personal_safety, damage, catch_condition, fishing_style.

Levels: no, yes.

Description: psychological aspect, elements such as anxiety and sense of fatigue which can eventually drives motivation (Chatterjee and and Bolar, 2019). The stressful events considered are having damages to the gear, being injured, being the catch in poor condition. The CPT is filled with strategy M2, where the answers to Q21 a, b and c (stress) are used as weights..

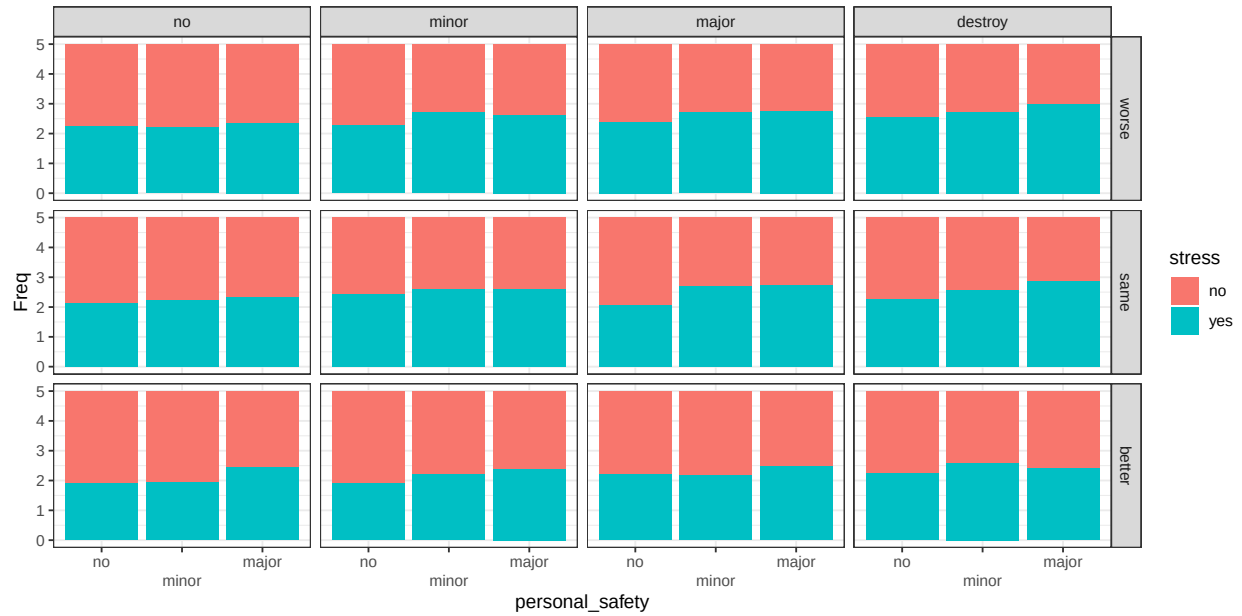


Figure A2.12: conditional distribution for stress . Columns represents levels of the node damage; rows are levels of the node catch_condition.

Catches

Parents: go_out, stock_status, catchability.

Levels: no, poor, average, good.

Description: relaised caught.

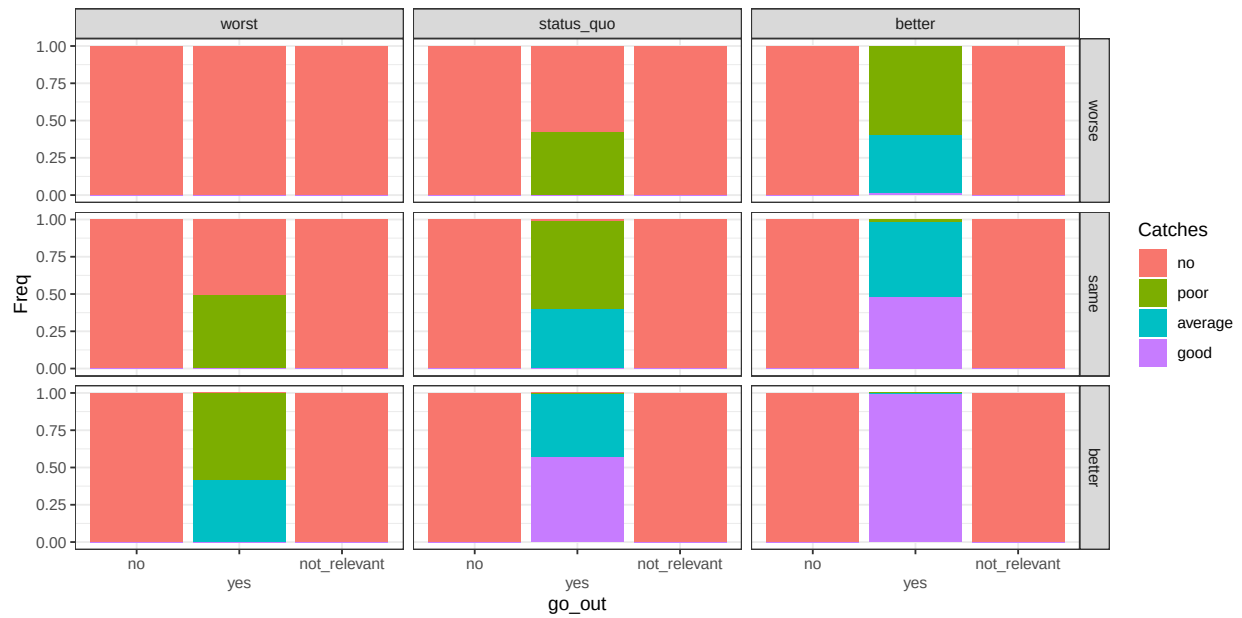


Figure A2.13: conditional distribution for Catches . Columns represents levels of the node stock_status; rows are levels of the node catchability.

Profit

Parents: Catches, cost.

Levels: negative, negligible, positive.

Description: realised profit.

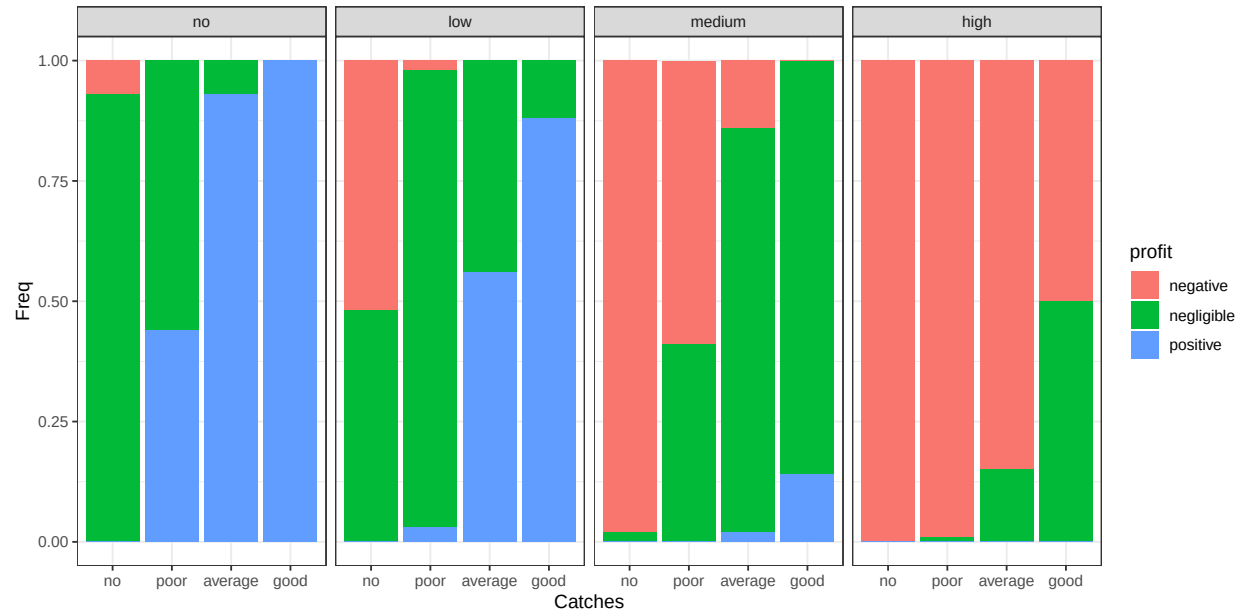


Figure A2.14: conditional distribution for profit . Columns represents levels of the node cost.

Health

Parents: stress, personal_safety.

Levels: Low, Medium, High.

Description: divorced node that summarizes variables related to physical and mental health. All the conditions are necessary to obtain a large value; therefore, we implement the weighted min algorithm (Eq.). Also, the contribution is uneven, with stress contributing 20% less than injury.

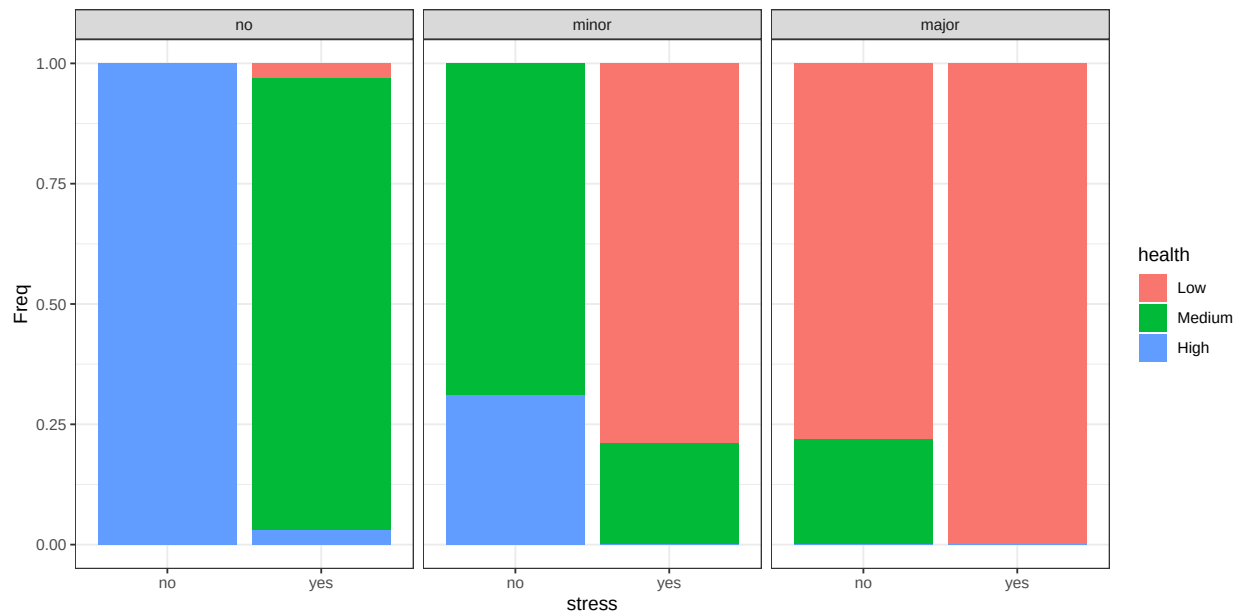


Figure A2.15: conditional distribution for health . Columns represents levels of the node personal_safety.

Satisfaction

Parents: Catches, catch_condition.

Levels: Low, Medium, High.

Description: Summarizes variables related to the non-monetary satisfaction derived from the fishing experience. This includes the time spent fishing, the catch amount and the average fish size. All the variables are calculated as relative change. All the conditions are necessary to obtain a large value; therefore, we implement the weighted min algorithm (Eq.). In this case we give equal weight.

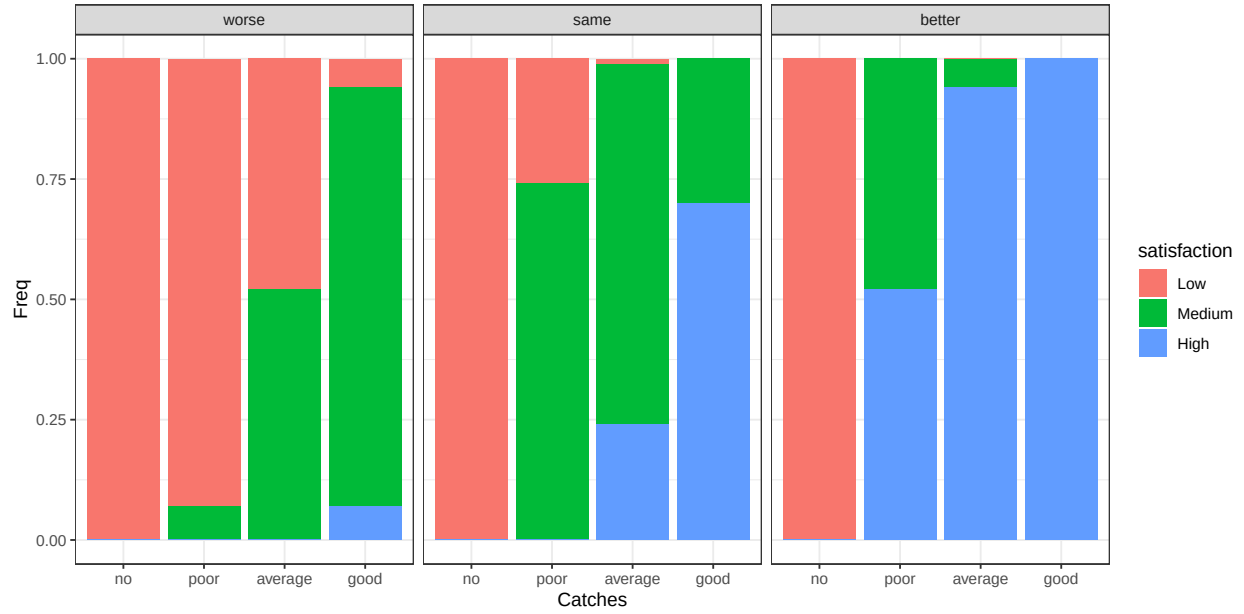


Figure A2.16: conditional distribution for satisfaction . Columns represents levels of the node catch_condition.

Cost

Parents: gear, additional_mitigation, damage.

Levels: no, low, medium, high.

Description: a summary of the extra expected price range to reach the fishing grounds, to implement maintenance (after or before the event to adapt to increased risk hazard), to replace the fishing gear.



Figure A2.17: conditional distribution for cost . Columns represents levels of the node additional_mitigation; rows are levels of the node damage.

Non_monetary_value

Parents: health, satisfaction, go_out.

Levels: Low, Medium, High.

Description: summarizes the outcomes that are not related with money. It depends on the possibility to go out fishing and to obtain satisfying catches, while staying safe. In this case all the conditions are necessary to obtain a large value, therefore we implement the weighted min algorithm (Eq.). The health node is Low-Medium-High; the satisfaction is below-average-more.

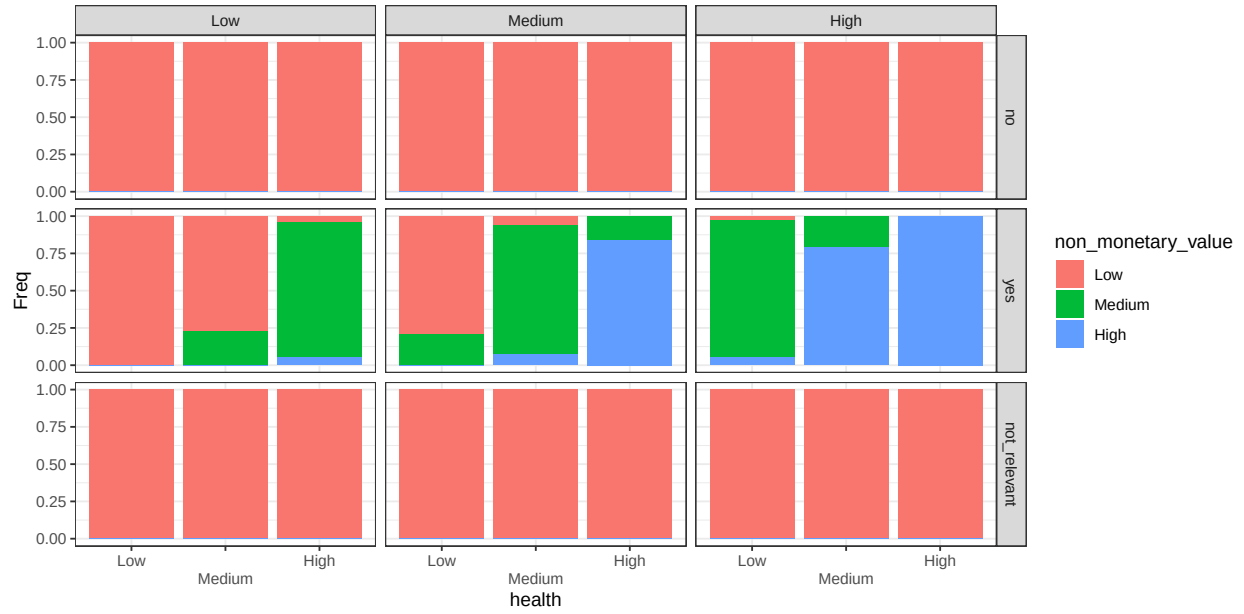


Figure A2.18: conditional distribution for non_monetary_value . Columns represents levels of the node satisfaction; rows are levels of the node go_out.

Credit

Parents: fishing_style.

Levels: very_difficult, difficult, neutral, easy, very_easy.

Description: The possibility to obtain loans for investments.

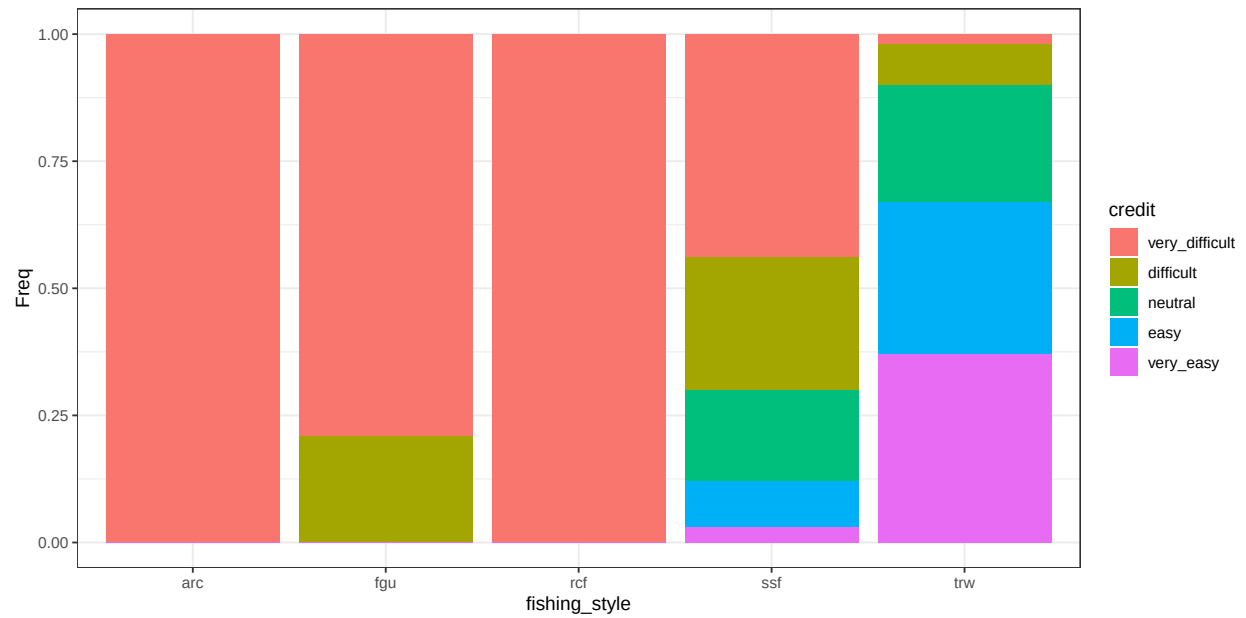


Figure A2.19: conditional distribution for credit .

Entrepreneurship

Parents: fishing_style.

Levels: low, medium, high.

Description: Motivation and capabilities of individuals to explore new technologies and methods for fishing and marketing.

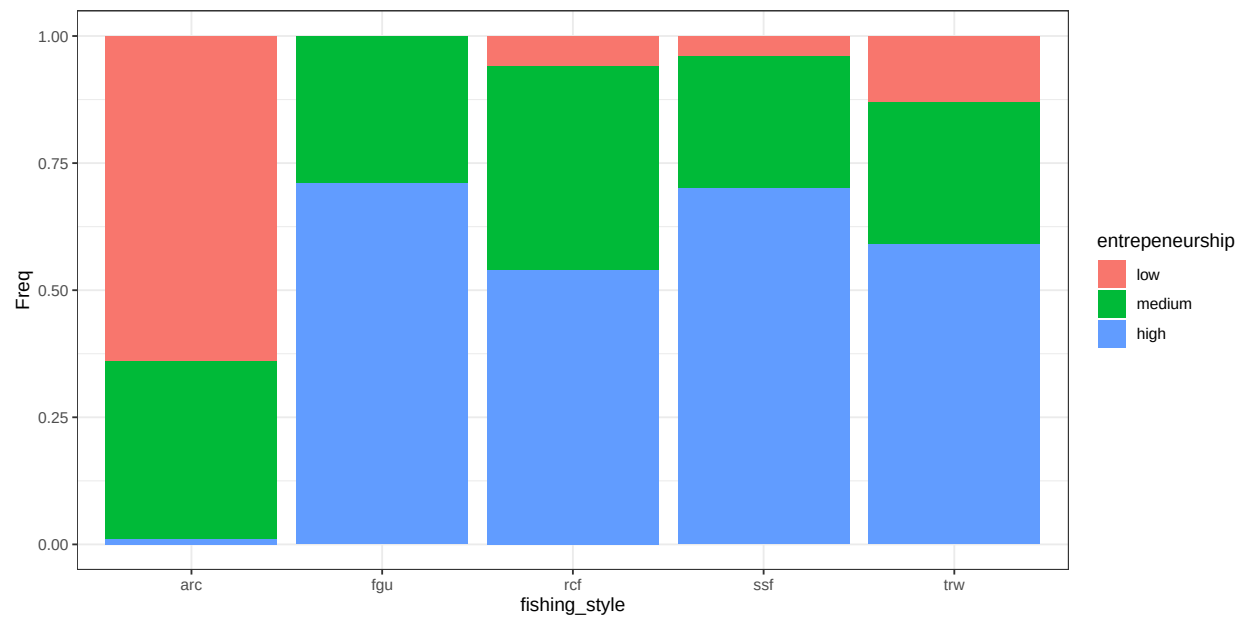


Figure A2.20: conditional distribution for entrepreneurship .

Innovative capacity

Parents: credit, entrepreneurship.

Levels: low, medium, high.

Description: The capacity to pursue different avenues (in fishing, marketing, technology, etc.) and to adapt to changing circumstances.

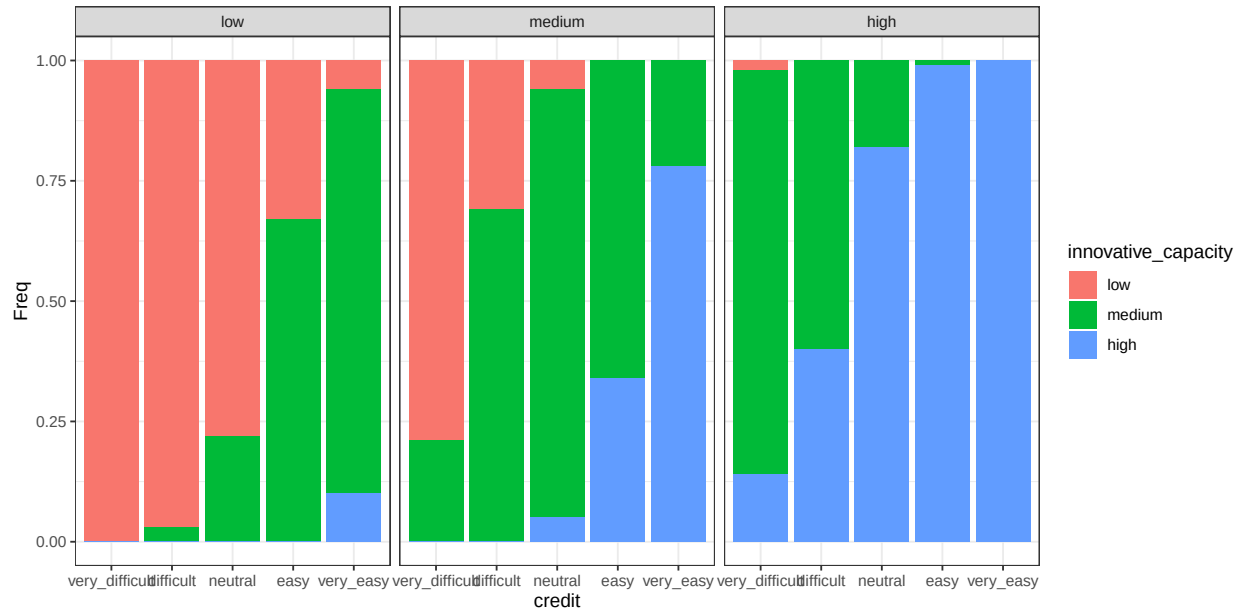


Figure A2.21: conditional distribution for innovative_capacity . Columns represents levels of the node entrepreneurship.

Professional

Parents: fishing_style.

Levels: No, Yes.

Description: measures how fishing is a source of income. Full-time job; side-job; recreational (or leisure).

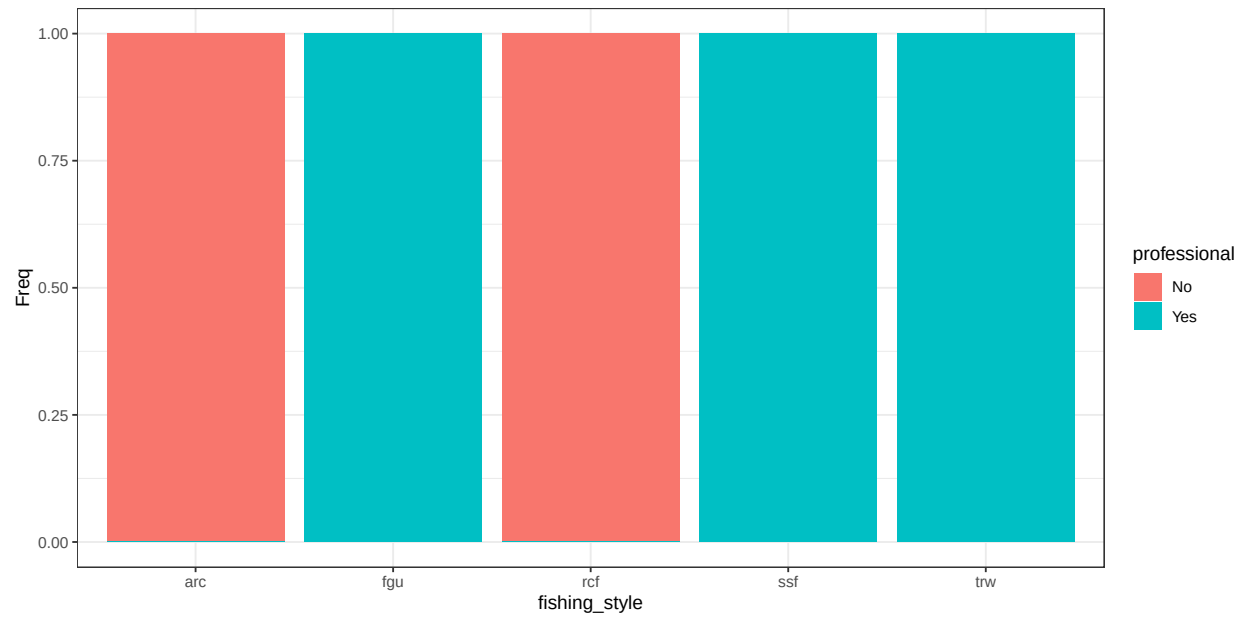


Figure A2.22: conditional distribution for professional .

Job mobility

Parents: fishing_style.

Levels: low, medium, high.

Description: The extent to which non-fishing income-generating activities are accessible to an individual or household.

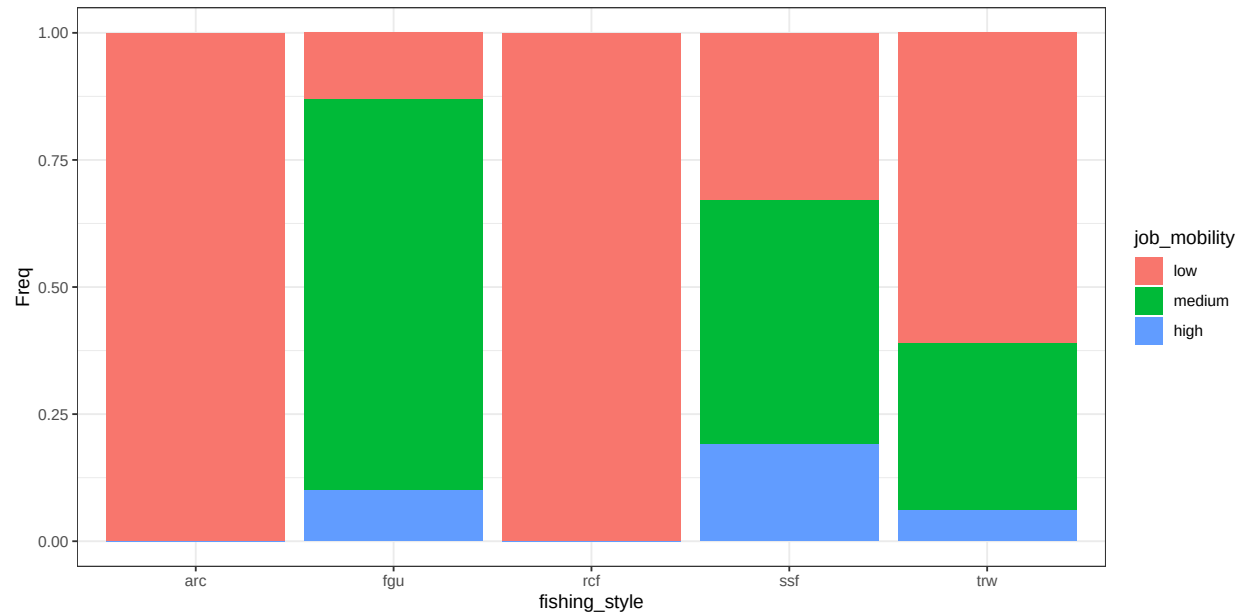


Figure A2.23: conditional distribution for job_mobility .

Economic importance

Parents: job_mobility, professional.

Levels: Not_important, Somehow_important, Very_important.

Description: Summarizes the importance of the human community dimensions (economic, societal, individual) to different fisherman.

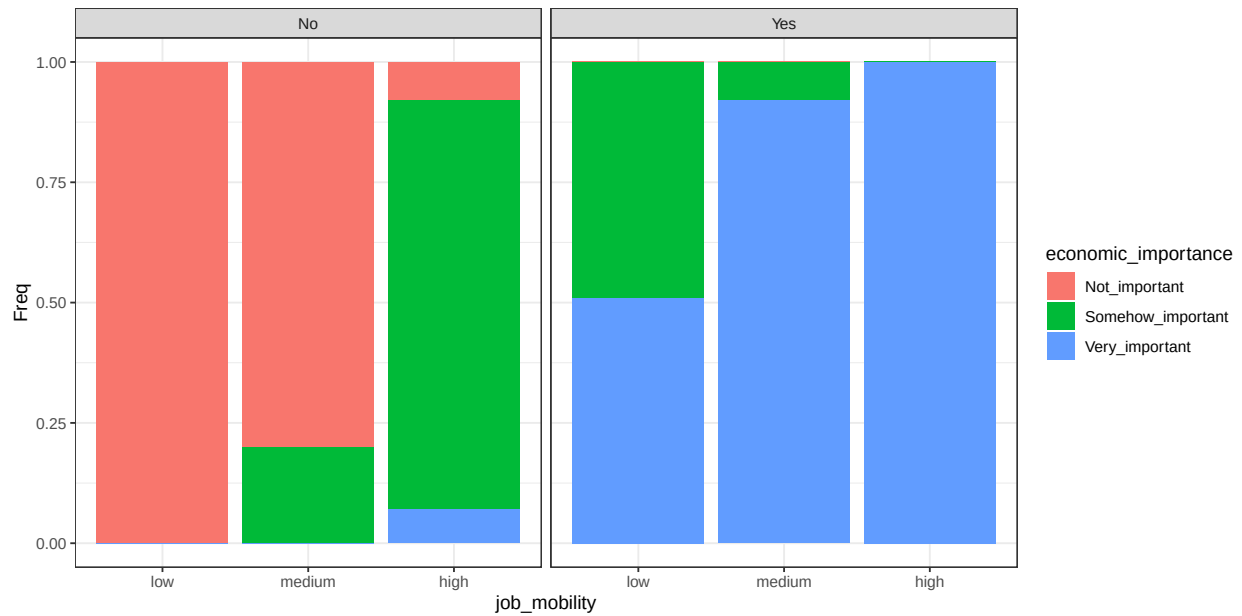


Figure A2.24: conditional distribution for economic_importance . Columns represents levels of the node professional.

Knowledge transmission

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: how important do you consider transmitting your fishing knowledge to future generations.

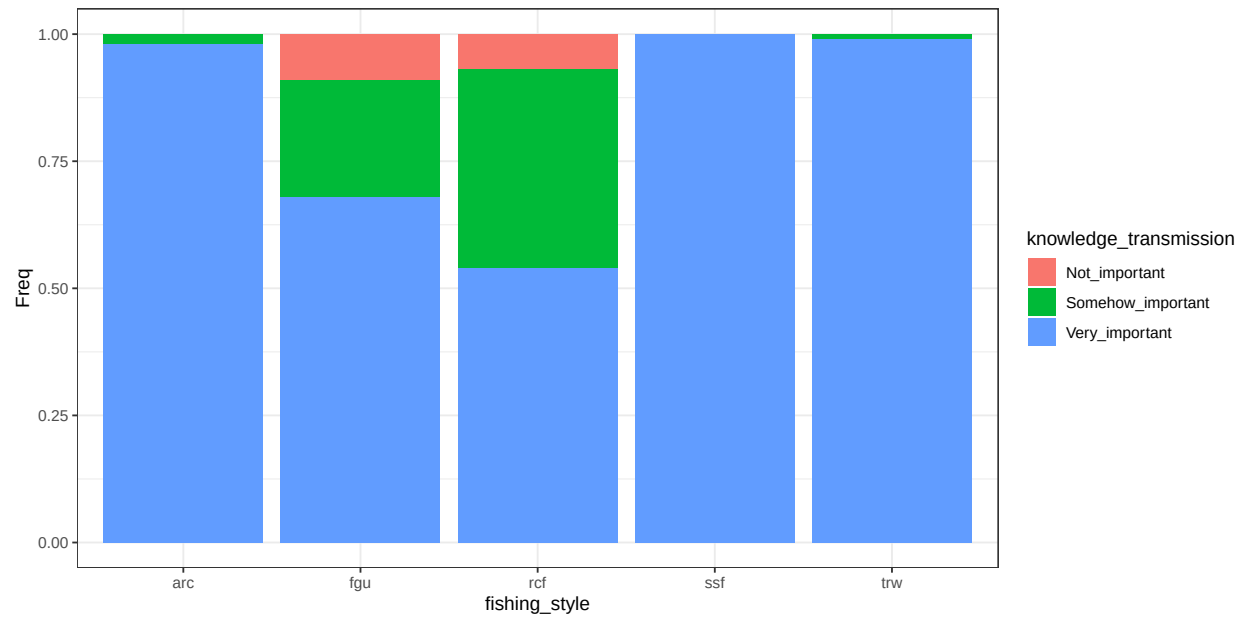


Figure A2.25: conditional distribution for knowledge_transmission .

Benefit local_community

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: how important is fishing to maintain social relations within your community.

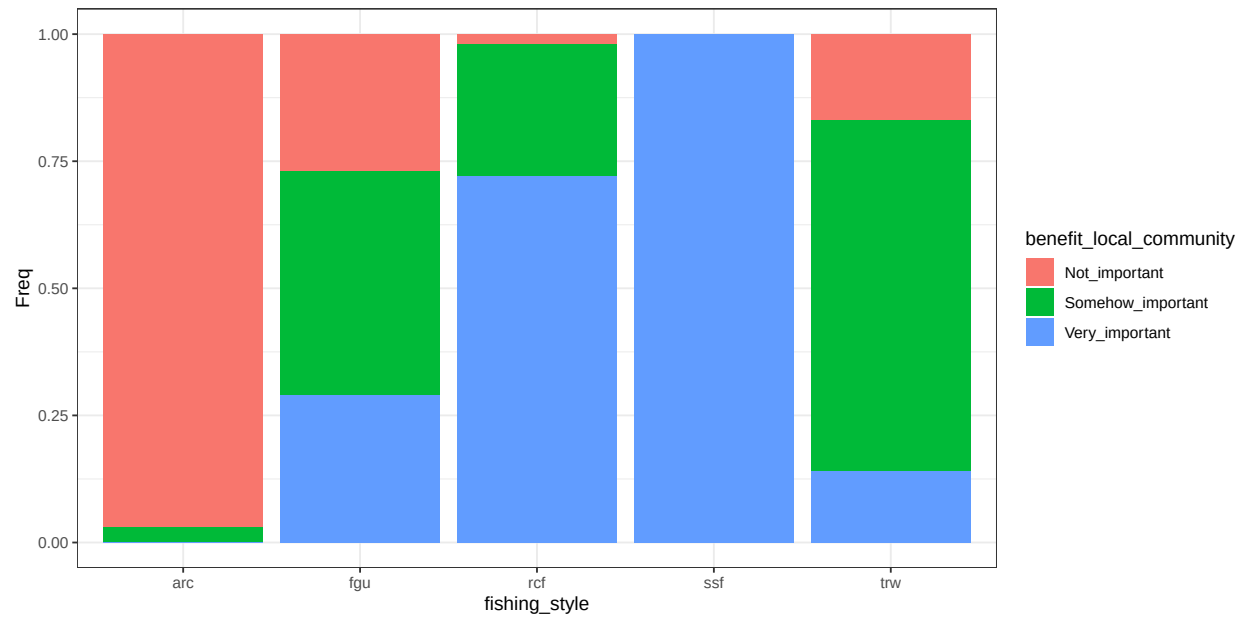


Figure A2.26: conditional distribution for benefit_local_community .

Traditional food

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: is your fishing activity contributing to provide for traditional food?.

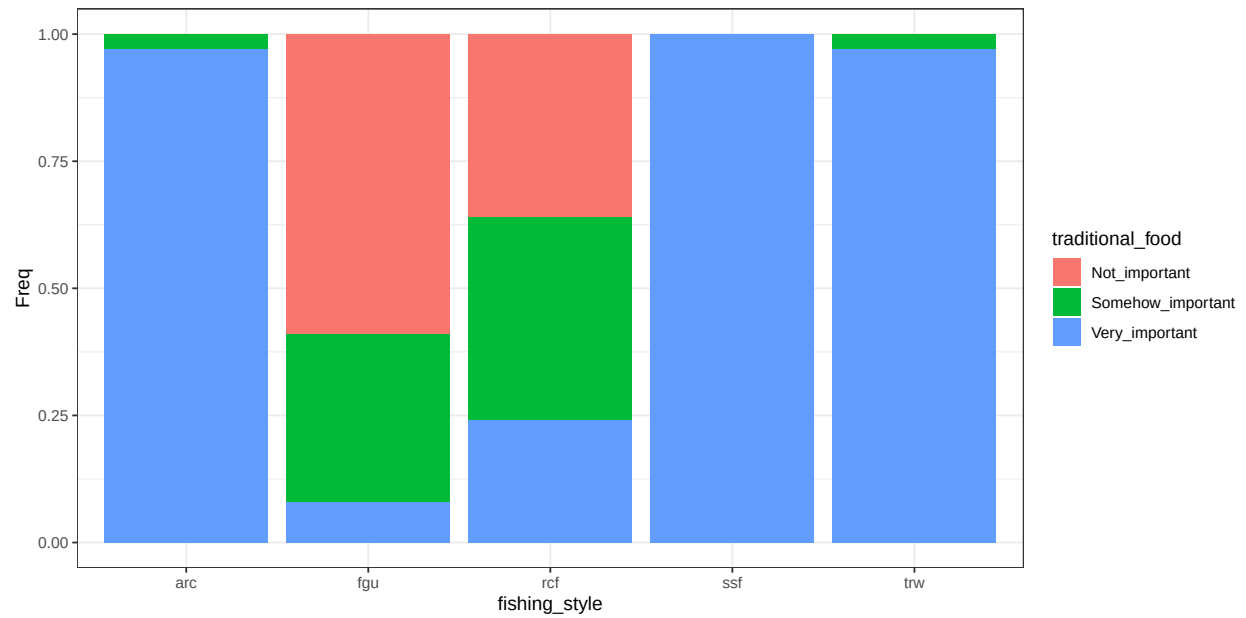


Figure A2.27: conditional distribution for traditional_food .

Benefit personal_community

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: how important is fishing to maintain social relations within your community.

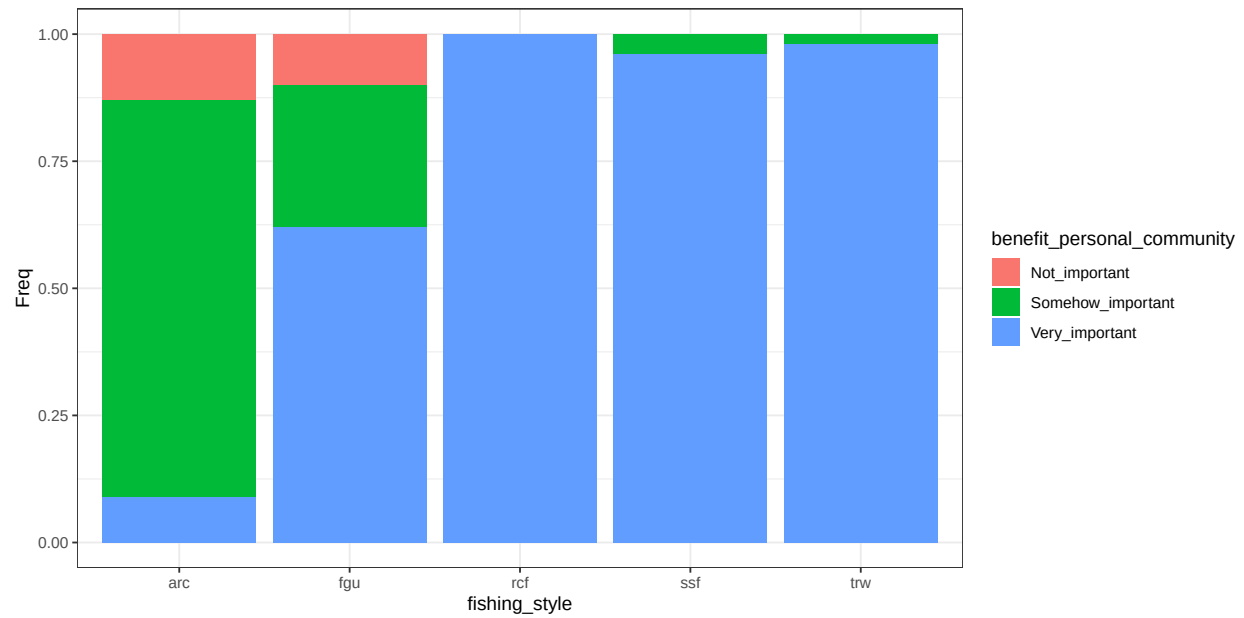


Figure A2.28: conditional distribution for benefit_personal_community .

Societal importance

Parents: knowledge_transmission, benefit_local_community, traditional_food, benefit_personal_community.

Levels: Not_important, Somehow_important, Very_important.

Description: Summarizes the importance of the human community dimensions (economic, societal, individual) to different fisherman.

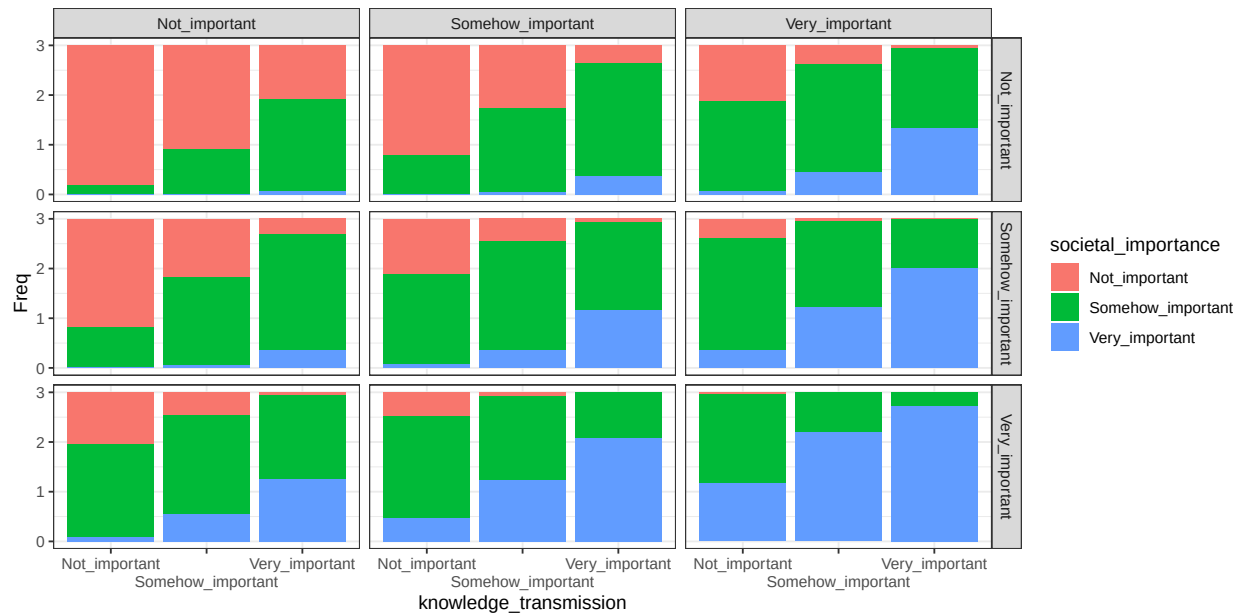


Figure A2.29: conditional distribution for societal_importance . Columns represents levels of the node benefit_local_community; rows are levels of the node traditional_food.

Identity

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: The degree to which fishing contributes to an individual's sense of identity, including self-perception.

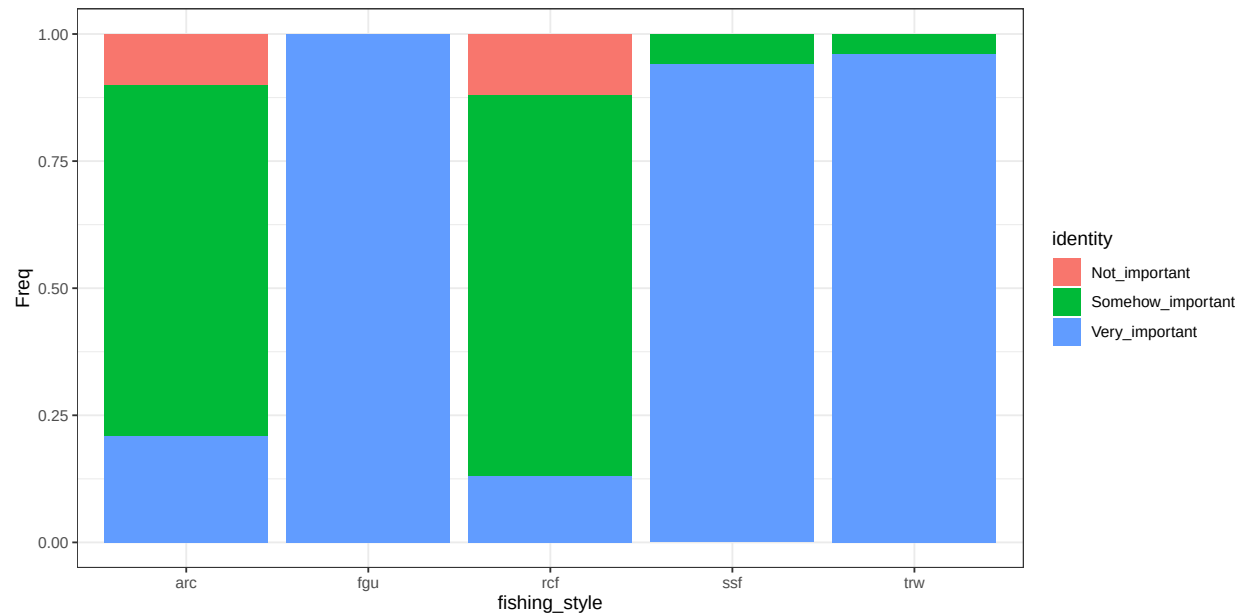


Figure A2.30: conditional distribution for identity .

Home

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: The degree to which fishing contributes to build a sense of place/home to a specific location. It can either be at sea or at land.

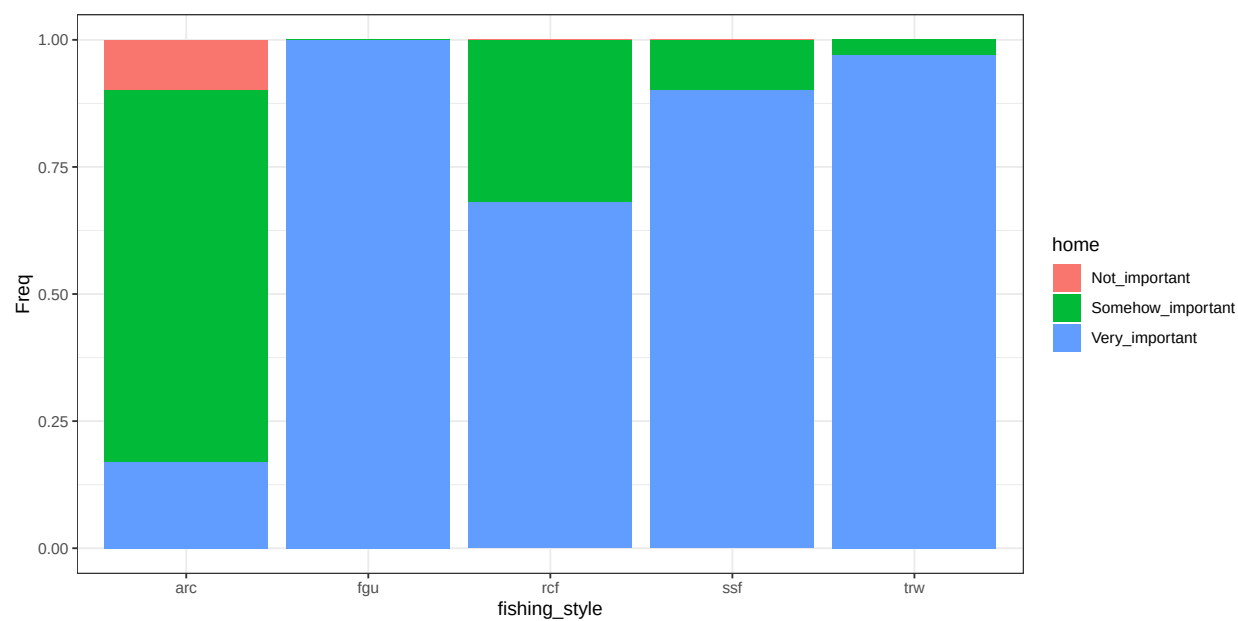


Figure A2.31: conditional distribution for home .

Outdoor benefit

Parents: fishing_style.

Levels: Not_important, Somehow_important, Very_important.

Description: The extent which an individual value the time spent outdoor (while fishing) to his personal well-being.

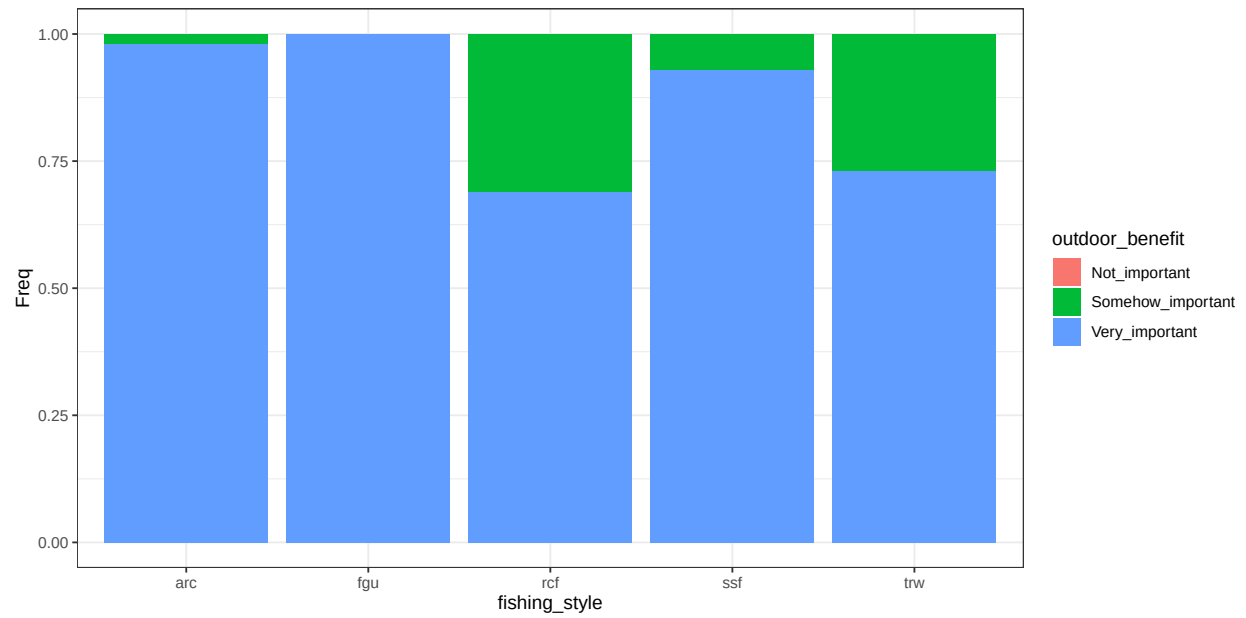


Figure A2.32: conditional distribution for outdoor_benefit .

Individual importance

Parents: outdoor_benefit, identity, home.

Levels: Not_important, Somehow_important, Very_important.

Description: Summarizes the importance of the human community dimensions (economic, societal, individual) to different fisherman.

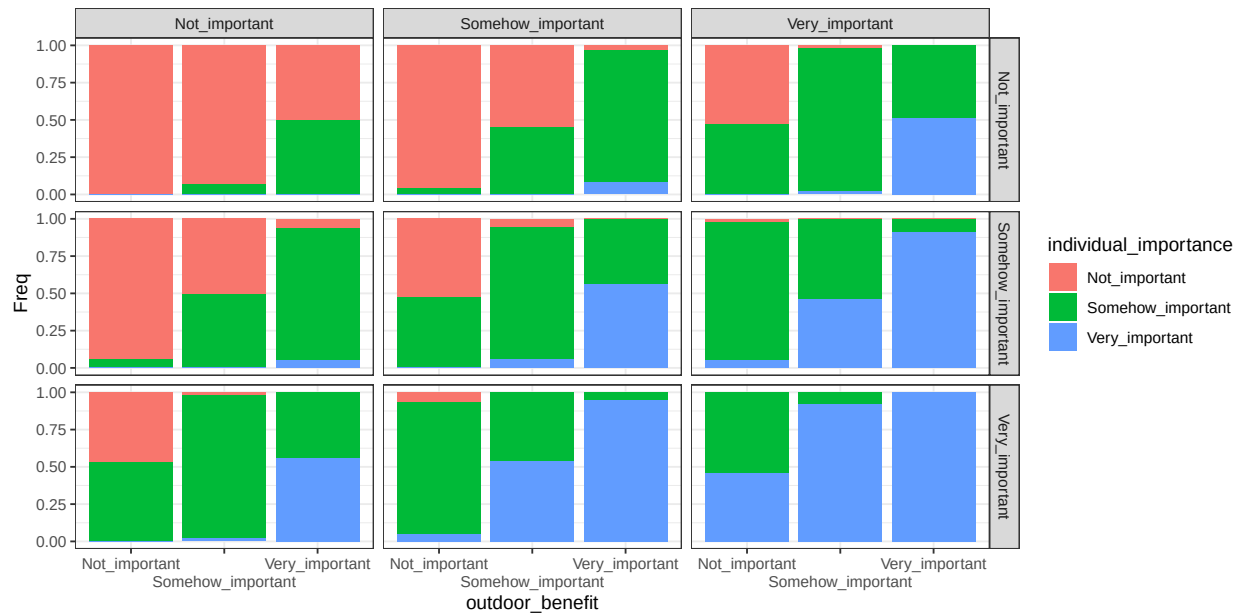


Figure A2.33: conditional distribution for individual_importance . Columns represents levels of the node identity; rows are levels of the node home.

Individual risk

Parents: individual_importance, non_monetary_value.

Levels: Low, Medium, High.

Description: aspects related to the individual well-being and satisfaction.

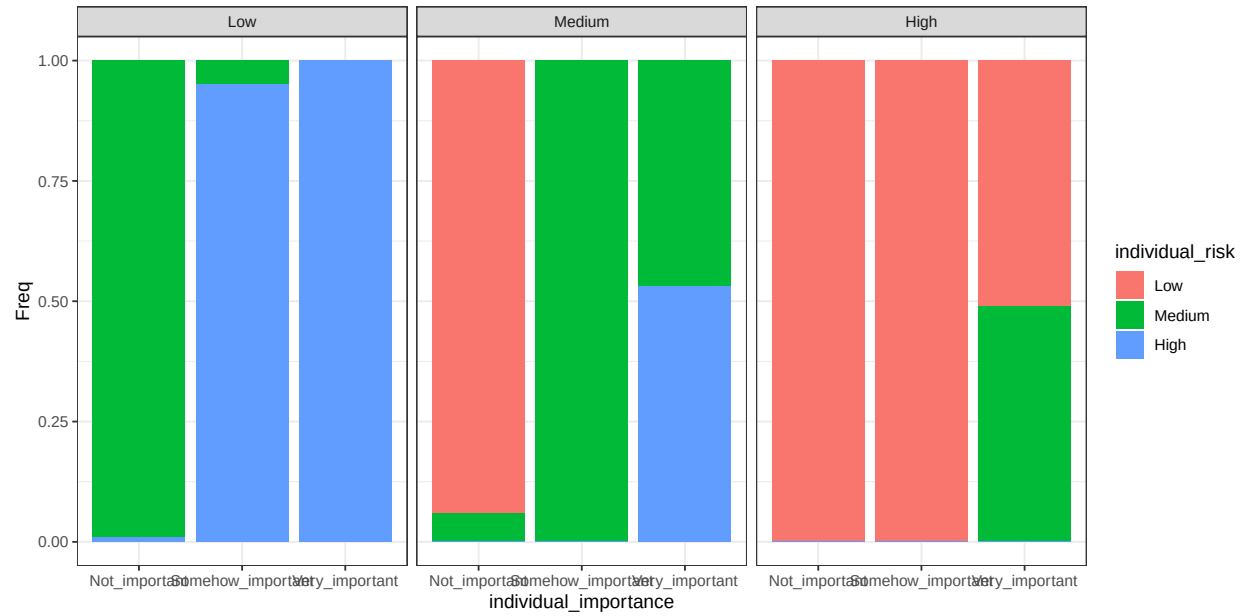


Figure A2.34: conditional distribution for `individual_risk` . Columns represents levels of the node `non_monetary_value`.

Societal risk

Parents: societal_importance, non_monetary_value.

Levels: Low, Medium, High.

Description: aspects related to the well-being of communities.

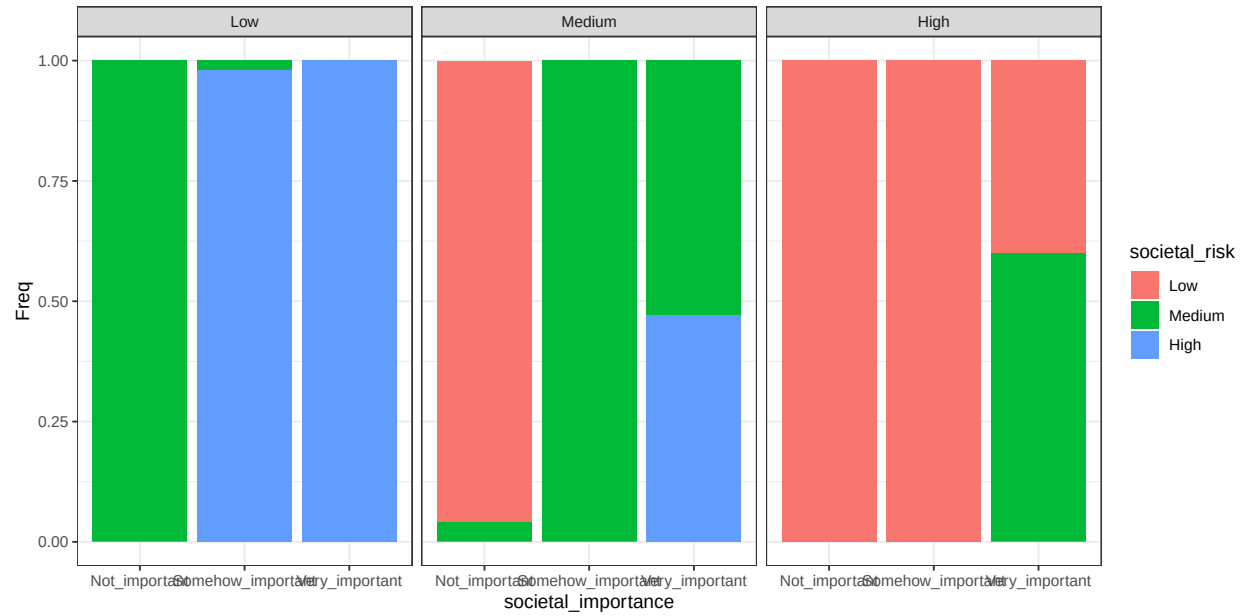


Figure A2.35: conditional distribution for `societal_risk` . Columns represents levels of the node `non_monetary_value`.

Economic risk

Parents: economic_importance, profit, innovative_capacity.

Levels: Low, Medium, High.

Description: economic aspects related to fishing practices.

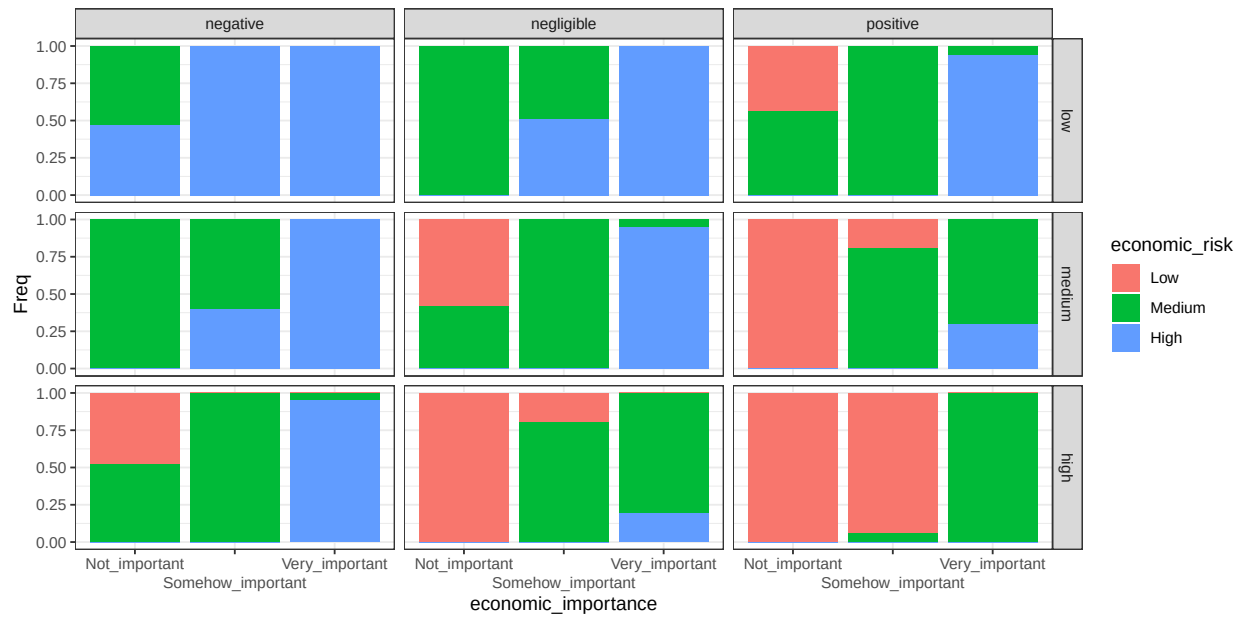


Figure A2.36: conditional distribution for *economic_risk* . Columns represents levels of the node *profit*; rows are levels of the node *innovative_capacity*.